

Assignment 4: Code Error Analysis Sheet

Comprehensive Analysis, Verification, and Debugging Log for Source Files (1.c to 5.c)

Source File: 1.c

Question Number	Error	Line Number	Error Type	Error Correction
1.c - Error 1	Missing line-terminating semicolon (;) after variable initialization block.	11	Syntax Error	Add a semicolon at the end of the statement: int department = 1;
1.c - Error 2	Type mismatch configuration; assigning a string literal directly to a primitive integer variable.	13	Type Mismatch	Change string to an integer value constant: int marks = 90;
1.c - Error 3	Using assignment operator (=) inside conditional statement instead of the equality relational check (==).	26	Logical Error	Change to equality operator expression: if(x == 10)
1.c - Error 4	Referencing an entirely undeclared variable symbol token name (score).	31	Undeclared Variable	Declare int score; before accessing or printing it.
1.c - Error 5	Invoking function add() without a forward prototype declaration visible before invocation.	53	Implicit Declaration Error	Add a prototype declaration signature for add above main() .
1.c - Error 6	Accessing localized variable marker temp outside its defined local curly braces block scope boundaries.	61	Scope Error	Relocate print inside scope block or declare variable at broader scope level.
1.c - Error 7	Array index out of bounds; attempting to access index offset 10 on an array sized for only 5 elements.	65	Array Bounds Error	Keep the indexing index bounded within safe dimensions: arr[4] = 100;
1.c - Error 8	Format specifier mismatch; utilizing integer token specifier flag (%d) to display float variables.	69	Format Specifier Mismatch	Correct format specifier to floating point presentation flag: printf("%f ", salary);
1.c - Error 9	Accessing localized loop counter parameter tracking index result outside the loop code body scope block.	89	Scope Error	Declare int result; outside/above the loop construct.
1.c - Error 10	Identical processing console tracking outputs written for both true and false paths of the conditional block.	93-96	Logical Error	Modify the output text inside the alternative else statement branch block.
1.c - Error 11	Calling function displayData() without passing its mandatory expected integer value argument.	100	Function Call Mismatch	Supply the required integer value variable parameter: displayData(value);
1.c - Error 12		125	Logical Error	

Question Number	Error	Line Number	Error Type	Error Correction
	Trailing semicolon placed directly behind while loop parameter block creates an empty infinite processing statement loop.			Remove the trailing semicolon to bind loop body statements block: while(z > 0) { z--; }
1.c - Error 13	Assigning double quote character string literal data straight to a single char variable type block primitive.	130	Type Mismatch	Convert string literal quotes to single character literal character quote notation: char grade = 'A';
1.c - Error 14	Passing a plain raw integer variable directly into a string address specifier checker (%s), causing runtime collapse.	134	Runtime Error (Crash)	Correct flag to match data width properties accurately: printf("%d ", number);
1.c - Error 15	Dereferencing a hardcoded NULL pointer reference tracking point address cell.	138	Pointer Error (SegFault)	Ensure target references point directly to active initialized memory objects.
1.c - Error 16	Invoking function calculateAverage() without any prior visible prototype definition layout rules.	142	Undeclared Function Call	Declare or define the function interface before executing main driver parameters.

Source File: 2.c

Question Number	Error	Line Number	Error Type	Error Correction
2.c - Error 1	Type mismatch; assigning a double-quoted string literal directly to primitive int variable.	10	Type Mismatch	Assign a valid numerical integer data literal value: int marks = 90;
2.c - Error 2	Missing trailing statement line-termination semicolon delimiter.	16	Syntax Error	Append the terminal separation mark character: printf("Program Started ");
2.c - Error 3	Attempting to extract absolute memory address reference coordinates of local CPU register class variable storage.	42	Semantic Mismatch	Remove register keyword storage designation parameter attribute.
2.c - Error 4	Accessing block-local loop helper variable parameter temp completely outside loop bracket scopes.	62	Scope Error	Declare indexing/accumulator trackers outside or above loop execution scopes.
2.c - Error 5	Critical active runtime execution division exception due to literal calculation division by zero denominator variables.	71	Runtime Error (Exception)	Add checks ensuring denominator data states are not zero prior to division.
2.c - Error 6	Reading and printing contents of completely uninitialized local data stack memory variable blocks.	123	Logical Error / Garbage Read	Provide explicit clear initial baseline data settings during declaration checks: int data = 0;

Source File: 3.c

Question Number	Error	Line Number	Error Type	Error Correction
3.c - Error 1	Missing ending structural line-separation semicolon character behind loop configuration choice primitive variable.	21	Syntax Error	Append ending delimiter mark: int choice;
3.c - Error 2	Missing statement block colon selector punctuation indicator behind switch statement catch-all default label branch.	55	Syntax Error	Append block label delimiter marker suffix symbol: default:
3.c - Error 3	Database overflow check validations trigger <i>*after*</i> dangerous array memory write storage steps execute.	81	Logical Sequence Error	Reorder criteria validations step flow to assert array allocation bounds <i>*prior*</i> to writes.
3.c - Error 4	Loop limit evaluations map across out of bounds off-by-one errors using invalid boundary conditions relational check parameters (<=).	90	Array Bounds Error	Invert relational expression flag elements to stay within boundaries: i < totalStudents
3.c - Error 5	Hardcoded literal evaluation path divides data fields directly by literal numeric value zero.	115	Runtime Fault Danger	Divide instead by active counter variables: return total / totalStudents;
3.c - Error 6	Referencing non-existent structural token instance label typo name missing context specifier tracks (totalStudent).	140	Undeclared Identifier	Correct typo character string to point to true active global tracking variable: totalStudents
3.c - Error 7	Assigning literal NULL strings directly onto entire static block structures fields variable parameters.	178	Semantic Type Mismatch	Clear elements individually or perform block clear macros manually.
3.c - Error 8	Comparing multi-byte character array contents references using basic identity equivalence check operators (==).	187	Logical Bug	Utilize explicit string content text match inspection utilities: if(strcmp(name, students[0].name) == 0)
3.c - Error 9	Opening persistent files using read-only access mode string characters tokens (" r ") while demanding stream writing tracks outputs.	211	File IO Access Bug	Alter stream access descriptors strings configuration tokens to write layouts: fopen("student.txt", "w");

Source File: 4.c

Question Number	Error	Line Number	Error Type	Error Correction
4.c - Error 1	Loop counter limits check triggers off-by-one boundary array out of bounds overflows utilizing (<=).	93	Array Bounds Error	Correct relational conditions boundaries flags parameters settings: i < totalAccounts
4.c - Error 2	Inverted business requirement conditional logic parameter triggers validation block errors backward on account transactions tasks.	136	Logical Code Flow Bug	Invert the direction of comparison validation operator flags layout strings: if(amount > accounts[acc].balance)
4.c - Error 3	Referencing non-existent structural token identifier instance name missing plural context attributes (totalAccount).	185	Undeclared Identifier	Correct spelling target string to match exact global declaration state: totalAccounts
4.c - Error 4	Assigning text values string literals directly onto primitive single integer struct field parameters cells.	195	Type Mismatch Mappings	Set values explicitly through valid primitive integer numeric constant literals.

Source File: 5.c

Question Number	Error	Line Number	Error Type	Error Correction
5.c - Error 1	Missing statement block choice token line separation terminal delimiter behind primitive menu input variables definitions.	21	Syntax Error	Append standard punctuation character: int choice;
5.c - Error 2	String duplication routines pass long input sequences straight into narrow array container stack frames causing buffer overflow.	257	Stack Buffer Overflow	Enlarge character destination array width properties to safely encapsulate text values data strings.
5.c - Error 3	Dynamic system heap allocation requests reserve tracking metrics blocks smaller than indexing usage operations.	266	Heap Corruption Trap	Scale up memory requests blocks count allocations multiplied by target properties types sizes metrics.
5.c - Error 4	Accessing localized system heap spaces and references following terminal memory deallocation sequences (Use-After-Free).	274	Pointer Lifecycle Error	Nullify active tracking reference address registers instantly following file or block clearance tasks.
5.c - Error 5	Double execution of clear memory routines triggers fatal memory architecture core crashes.	276	Double Free Violation	Remove the duplicate secondary line clearing parameters statement block completely.